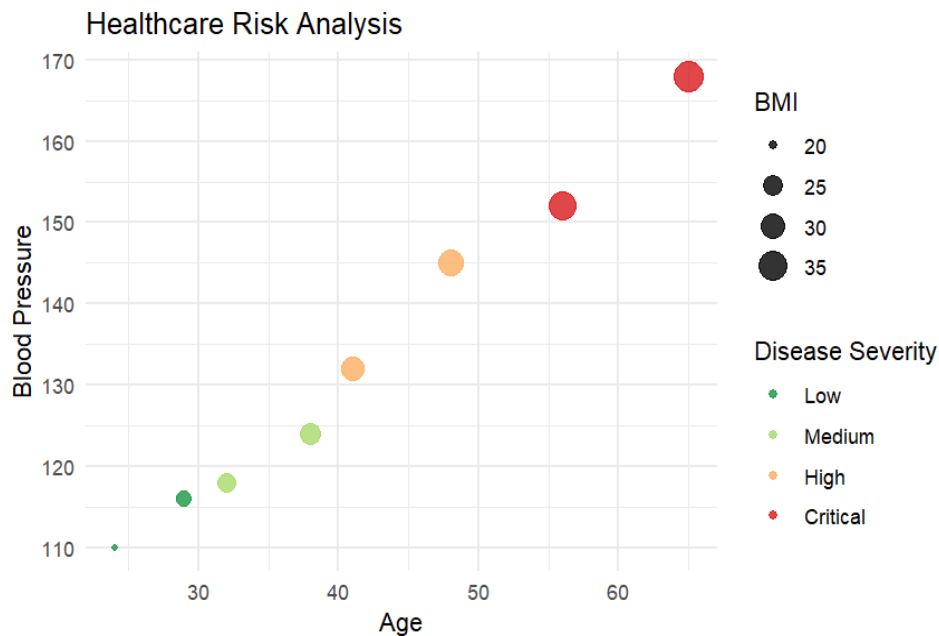


CO2 – Assessment Test 1 (AT1)

Chart Construction Test

Question 1 – Healthcare Analytics



Code

```
# Install package (if not already installed)
install.packages("ggplot2")

# Load library
library(ggplot2)

# Create dataset
health <- data.frame(
  Patient = c("P1", "P2", "P3", "P4", "P5", "P6", "P7", "P8"),
  Age = c(24, 32, 41, 48, 56, 65, 38, 29),
  BMI = c(19.8, 24.5, 28.1, 30.3, 33.2, 36.5, 26.2, 22.1),
  BloodPressure = c(110, 118, 132, 145, 152, 168, 124, 116),
  Severity = factor(
    c("Low", "Medium", "High", "High", "Critical", "Critical", "Medium", "Low"),
    levels = c("Low", "Medium", "High", "Critical")
  )
)

# Scatter Plot
```

```

ggplot(
  health,
  aes(
    x = Age,
    y = BloodPressure,
    color = Severity,
    size = BMI
  )
) +
  geom_point(alpha = 0.8) +
  scale_color_brewer(palette = "RdYlGn", direction = -1) +
  labs(
    title = "Healthcare Risk Analysis",
    x = "Age",
    y = "Blood Pressure",
    color = "Disease Severity",
    size = "BMI"
  ) +
  theme_minimal()

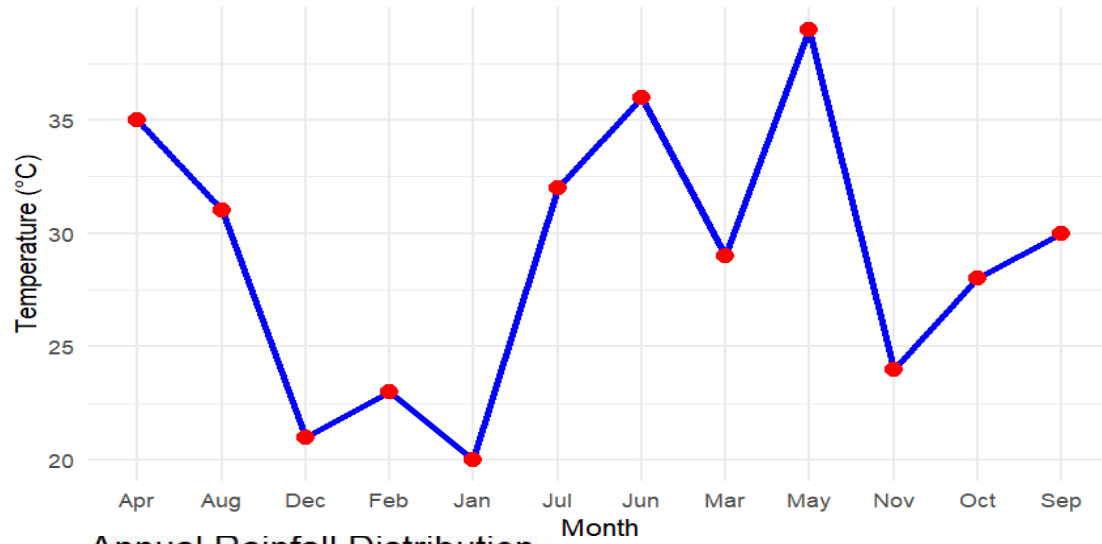
```

Justification

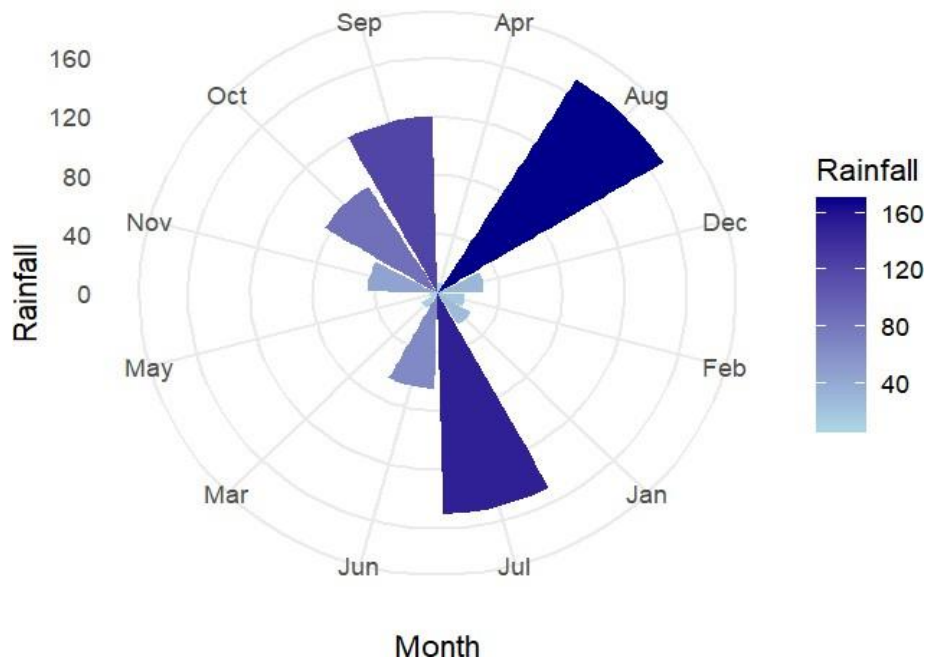
- Age (X-axis): Represents the age of each patient, helping identify trends in health risk across different age groups.
- Blood Pressure (Y-axis): Displays blood pressure levels, allowing comparison of cardiovascular health among patients.
- Disease Severity (Color): Different colors distinguish severity levels, making it easy to identify patients with low to critical health conditions.
- BMI (Point Size): Larger points represent higher BMI values, highlighting patients with greater obesity-related health risks.
- Visualization Benefit: The scatter plot combines four healthcare variables in a single chart, making it easier to analyze relationships between age, blood pressure, BMI, and disease severity.

Question 2 – Climate Science

Monthly Temperature Variation



Annual Rainfall Distribution



Code:

```
# Load library
library(ggplot2)

# Create dataset
climate <- data.frame(
  Month = c("Jan", "Feb", "Mar", "Apr", "May", "Jun",
            "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"),
  Solar = c(180, 200, 240, 280, 320, 300, 260, 250, 240, 220, 190, 170),
  Temperature = c(20, 23, 29, 35, 39, 36, 32, 31, 30, 28, 24, 21),
  Rainfall = c(25, 18, 12, 8, 5, 65, 150, 170, 120, 85, 45, 30)
)

# Cartesian Line Chart
ggplot(climate, aes(x = Month, y = Temperature, group = 1)) +
  geom_line(color = "blue", linewidth = 1.2) +
  geom_point(size = 3, color = "red") +
  labs(
    title = "Monthly Temperature Variation",
    x = "Month",
    y = "Temperature (°C)"
  ) +
  theme_minimal()

# Polar Bar Chart
ggplot(climate, aes(x = Month, y = Rainfall, fill = Rainfall)) +
  geom_bar(stat = "identity") +
  coord_polar() +
  scale_fill_gradient(
    low = "lightblue",
    high = "darkblue"
  ) +
  labs(title = "Annual Rainfall Distribution") +
  theme_minimal()
```

Comparison of Cartesian and Polar Charts**Cartesian Chart (Line Chart)****Advantages**

- Clearly displays monthly temperature changes over time.
- Makes it easy to identify increasing or decreasing trends.
- Suitable for continuous numerical data.
- Allows accurate comparison of temperature values between months.

Limitations

- Does not effectively highlight the cyclical nature of seasonal changes.
- Less visually appealing for representing annual climate cycles.

Polar Chart (Circular Bar Chart)**Advantages**

- Effectively represents seasonal rainfall patterns.
- Circular layout reflects the repeating annual climate cycle.
- Provides an attractive and engaging visualization.

Limitations

- Difficult to compare exact rainfall values across months.

- Less precise than Cartesian charts for quantitative analysis.
- Differences between bars may be harder to interpret due to the circular layout.

Question 3 – Business Analytics & Marketing

Code:

```
# Load library
library(ggplot2)

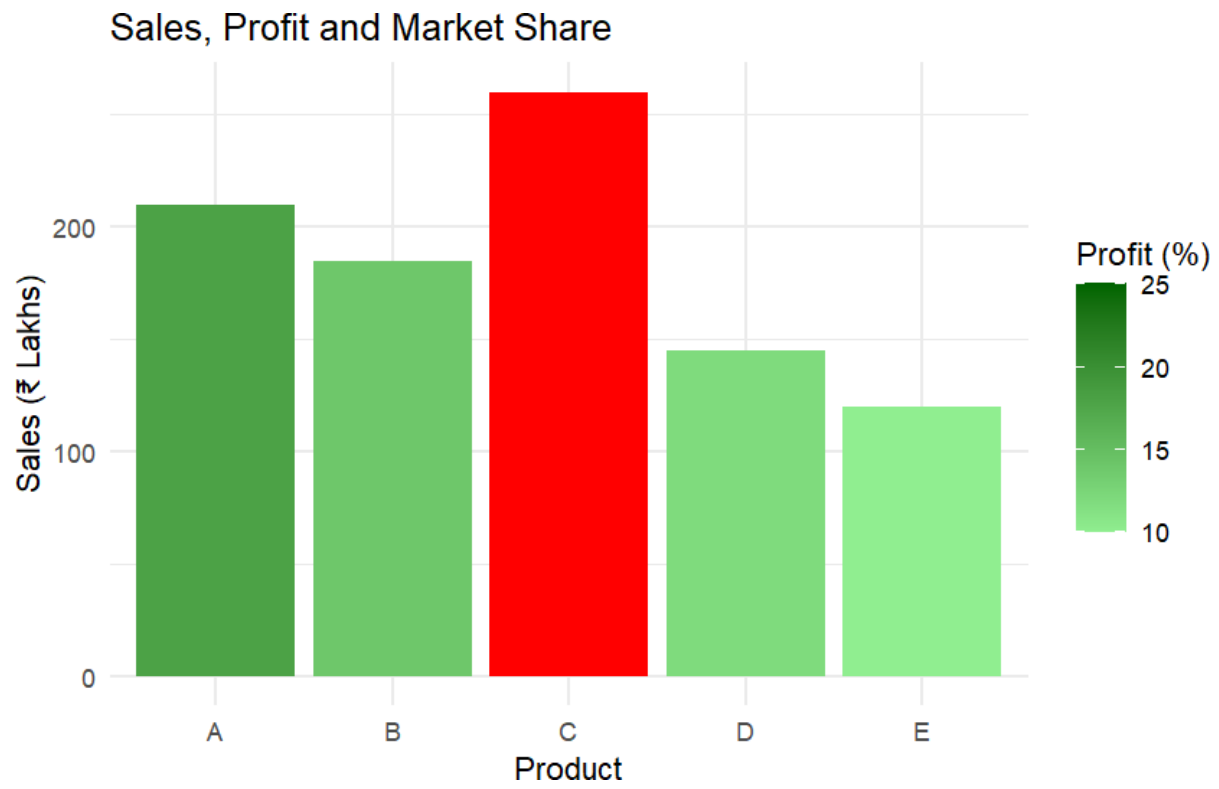
# Create dataset
business <- data.frame(
  Product = c("A", "B", "C", "D", "E"),
  Sales = c(210, 185, 260, 145, 120),
  Profit = c(18, 14, 25, 12, 10),
  MarketShare = c(24, 20, 28, 15, 13)
)

# Highlight product with the highest market share
business$Highlight <- ifelse(
  business$MarketShare == max(business$MarketShare),
  "Highest",
  "Others"
)

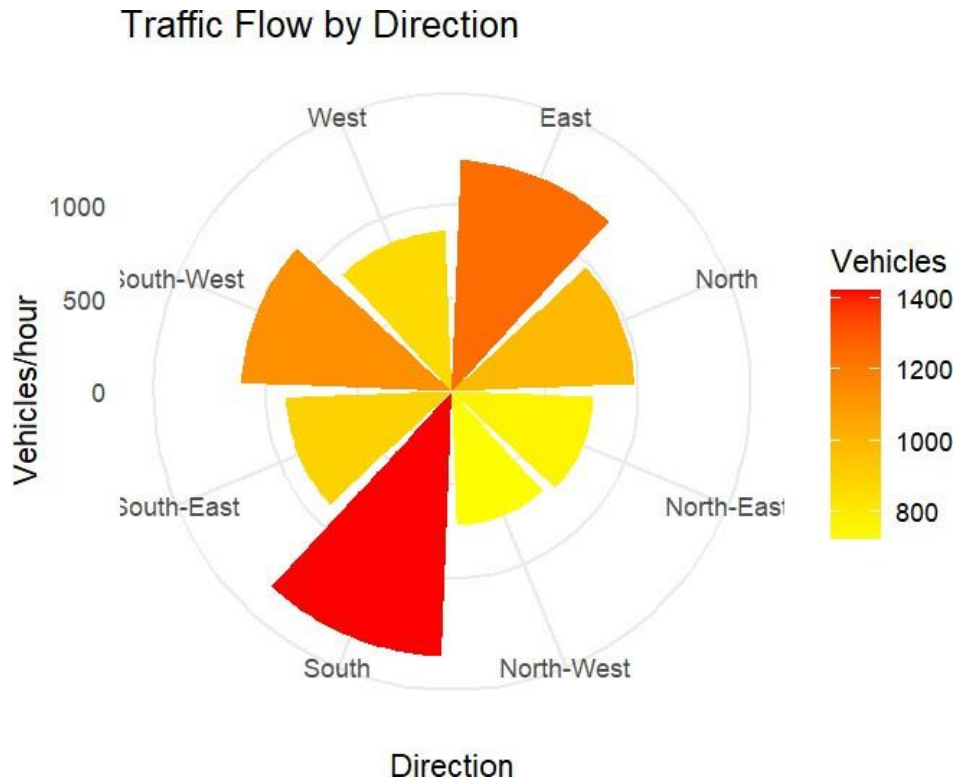
# Bar Chart
ggplot(business, aes(x = Product, y = Sales, fill = Profit)) +
  geom_bar(stat = "identity") +
  geom_bar(
    data = subset(business, Highlight == "Highest"),
    fill = "red",
    stat = "identity"
  ) +
  scale_fill_gradient(
    low = "lightgreen",
    high = "darkgreen"
  ) +
  labs(
    title = "Sales, Profit and Market Share",
    x = "Product",
    y = "Sales (₹ Lakhs)",
    fill = "Profit (%)"
  ) +
  theme_minimal()
```

Justification

- Bar Height (Sales): Represents the total sales generated by each product, making it easy to compare sales performance.
- Green Color Gradient (Profit): Uses a sequential green palette to indicate profit levels, where darker shades represent higher profit percentages.
- Red Highlight: Highlights the product with the highest market share, allowing quick identification of the market leader.
- Visualization Benefit: Combines sales, profit, and market share in a single chart, helping managers quickly identify the best-performing product and make informed business decisions.



Question 4 – Transportation Engineering



Code:

```
# Load library
library(ggplot2)

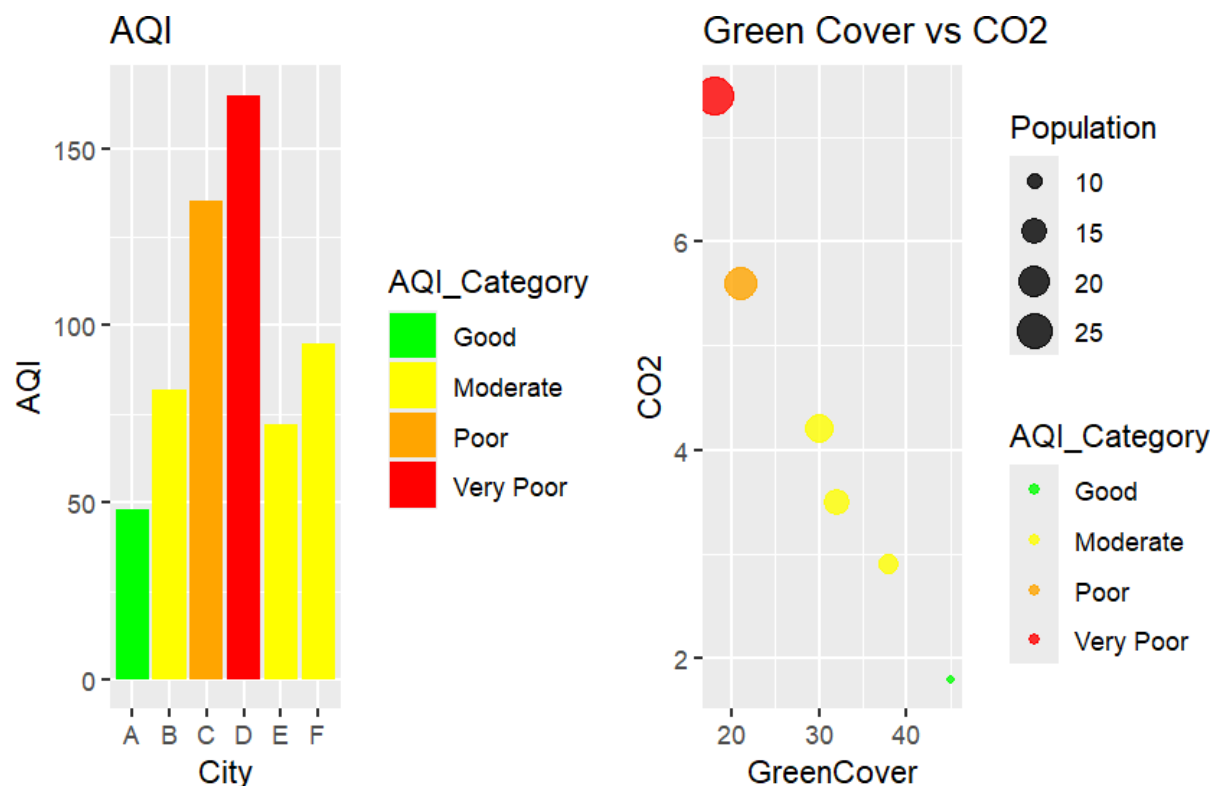
# Create dataset
traffic <- data.frame(
  Direction = c(
    "North", "North-East", "East", "South-East",
    "South", "South-West", "West", "North-West"
  ),
  Vehicles = c(980, 760, 1240, 890, 1420, 1130, 860, 720)
)

# Polar Bar Chart
ggplot(traffic, aes(x = Direction, y = Vehicles, fill = Vehicles)) +
  geom_bar(stat = "identity") +
  coord_polar() +
  scale_fill_gradient(
    low = "yellow",
    high = "red"
  ) +
  labs(
    title = "Traffic Flow by Direction",
    x = "Direction",
    y = "Vehicles per Hour"
  ) +
  theme_minimal()
```


Justification

- **Polar Layout:** The circular chart naturally represents compass directions, making it ideal for visualizing directional traffic flow.
- **Bar Length:** The length of each bar indicates the number of vehicles traveling in each direction, allowing easy comparison.
- **Color Gradient:** A yellow-to-red gradient represents traffic density, where darker red indicates higher vehicle volume and greater congestion.
- **Visualization Benefit:** The polar chart makes it easy to identify heavily congested directions and provides a more intuitive representation of directional traffic patterns than a traditional Cartesian chart.

Question 5 – Environmental Sustainability Dashboard



Code:

```
# Load libraries
library(ggplot2)
library(gridExtra)

# Create dataset
env <- data.frame(
  City = c("A", "B", "C", "D", "E", "F"),
  AQI = c(48, 82, 135, 165, 72, 95),
```

```

GreenCover = c(45, 32, 21, 18, 38, 30),
CO2 = c(1.8, 3.5, 5.6, 7.4, 2.9, 4.2),
Population = c(9, 15, 22, 28, 12, 18)
)

# Categorize AQI
env$AQI_Category <- cut(
  env$AQI,
  breaks = c(0, 50, 100, 150, 200),
  labels = c("Good", "Moderate", "Poor", "Very Poor")
)

# Plot 1: AQI by City
p1 <- ggplot(env, aes(x = City, y = AQI, fill = AQI_Category)) +
  geom_bar(stat = "identity") +
  scale_fill_manual(
    values = c(
      "green",
      "yellow",
      "orange",
      "red"
    )
  ) +
  labs(
    title = "Air Quality Index (AQI)",
    x = "City",
    y = "AQI"
  )

# Plot 2: Green Cover vs CO2
p2 <- ggplot(
  env,
  aes(
    x = GreenCover,
    y = CO2,
    size = Population,
    color = AQI_Category
  )
) +
  geom_point(alpha = 0.8) +
  scale_color_manual(
    values = c(
      "green",

```

```

    "yellow",
    "orange",
    "red"
)
) +
labs(
  title = "Green Cover vs CO2 Emissions",
  x = "Green Cover (%)",
  y = "CO2 Emissions"
)

# Display Dashboard
grid.arrange(p1, p2, ncol = 2)

```

Justification

- **Bar Chart:** Compares the Air Quality Index (AQI) across different cities, making it easy to identify cities with better or worse air quality.
- **Scatter Plot:** Illustrates the relationship between green cover and CO₂ emissions, helping to analyze how vegetation may influence pollution levels.
- **Point Size:** Represents the population of each city, allowing larger cities to be identified at a glance.
- **Color Scheme:** Uses a meaningful color palette to represent AQI categories:
 - **Green:** Good
 - **Yellow:** Moderate
 - **Orange:** Poor
 - **Red:** Very Poor